

Bioinformatics: Genomes and Biological Databases

Prof. David Jones
d.t.jones@ucl.ac.uk

The Biological Data Revolution

- Modern molecular biology is now being revolutionised by the availability of massive amounts of data, in particular sequence data
- New technologies: biological arrays, high-speed automated DNA sequencing equipment, laboratory robotics
- Genome Projects (100,000 Genomes etc.)
- Bulk sequencing of mRNA/cDNAs
- Information scientists have had a tough time keeping up with the data and it's getting harder

The raw product...

Start of Human Chromosome1

GAGTAGGTGGGATTACAGGCGTGCGCCACCATGCCCCGTCTAATTTTTGTATTTTCAGTAG
AGACGCGGTTTCGCCATCTTGGCCAGGCTGGTCTTGGAACCTCTGACCTCAAGTGATCTG
CCCGCCTCGGCCTCCCAAAGTGCTGGGATTACAGGCATGAGCCACTGTGCCCTGCCTAGC
TCACTATCTTTCAATCAGTAGAGATTCTTTAGTTATTTTTTAACTCCATGGATCCCAAGC
TTTGATTTTTGTGTTTCCAAACAAATTGCATTTATAAATAATAATTTTTTATTTATAATCAA
CAGACATCTAGGCTTGCTGTCAAGGCTTCTGATCAACATGAGATGACCGCCGTGTGGTAA
ACTGATGAACCCTGACCCATTAGGCTTTGGCTACAGAATGTGGAAATAAGTTGTGTTACT
ACATGTGTGTAATCCTAGGGTGCAGGACACCGGCCGGGAGGTTCCATAGAGTGATGGGTT
CTGCAGGTAACCTCATCCTCTAGTCCTCTGTAAGCTCCTAGAAGGAAGAAATTATGTCCTT
TAGACTAATAAAATTCTCCAAACCAAATACAGCACCTACTGTGAAGACACAAAGATACT
TTTAGAATAGTAAAACTTTATCCATTGAGAAATTCCTTAATGAAACAGTATCCAAGAAG
TCATTTGCCAGCAGATTTCTTAGAGGTGCGATAAAGAAGAGGACATTGCCAGTCGTCACA
GCAGCTGCAATAGCTCCTCTCTATTGTTAAACAGTGGGATATCTTGTGCAGGTTTTTCAGT
TGACAATCAATTTTAAAGATTAGTTTCGGTCCCCATCAATCAATTATTTATTAACCCATC
AATAAAAATTTAAATGCTCTGTGAGGTACAATAGCTATTAAAAGAGACAGAGGCACTTTC

... approximately 3 billion letters later ...

TAAACCACAATGACTCATTACTTCTCTTTGTTACTATTGGGAATCAGAGACATAGATTTT
GTTGATATTAGTCATTCAAATGAAATAAGCATGAATGTGCATACATTGGCTTTGTTTTCC
AAGGAGCTAACTTTTGGATGCAATAGCAATTTAATGAAAATTTCTTAGAGAATAACATGA
TACTTCAAACCAGACTATTTTAGAAACAAGAATAATGTTGAATTC

(End of Chromosome Y)

The Reference Human Genome

- Raw data collected: 2×10^{10} bases (characters ATGC) of *overlapping fragments*
- “Golden Path”: 3×10^9 bases (characters ATGC)
- 6×10^9 bits of information = 700 Mb (2 bits per base)
- Compresses (e.g. with zip) to ~ 680 Mb (~ 1.8 bits per base)
- The *differences* between an *individual's* genome and the standard reference genome compresses to around **4 Mb**

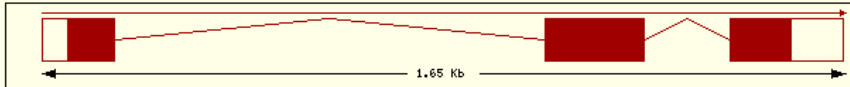
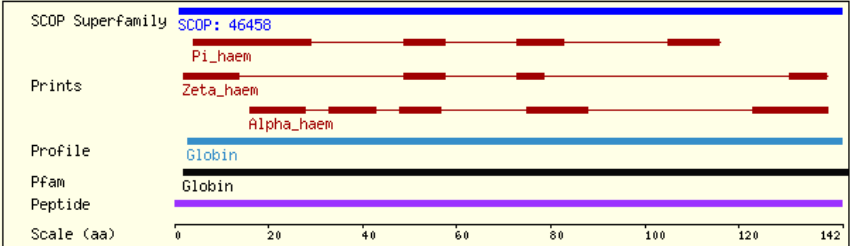
Genome Analysis

- Assembly of genomes
 - Reasonable hashing-based algorithms available
 - Still in progress (e.g. in metagenomics), but largely done
- Gene finding
 - Still work in progress: better algorithms still required.
 - Comparative techniques work well, however.
- Functional Annotation
 - Difficulty in keeping-up with data flow
 - Difficulty in assessing reliability
 - Standard algorithms limited
 - Ongoing - will continue for a long time!

Genome Annotation

- INPUT: Genome sequence data
- INPUT: Primary and secondary databases
- ALGORITHMS (mostly working on proteins)
 - Pattern Recognition
 - Structure Prediction
 - Text mining
- OUTPUT: Annotated genome sequences

Transcripts/Translation Summary

HBZ	Exons: 3 Transcript length: 589 bp Translation length: 142 residues [View transcript information] [View exon information] [View protein information]
Similarity Matches	This Ensembl entry corresponds to the following database identifiers: Affymx Microarray U133: HG-U133A:206647_at Affymx Microarray U95: HG-U95A:37217_at EMBL: J00182 [align] M24173 [align] Z84721 [align] HUGO: Search GeneCards for HBZ LocusLink: 3050 [align] MIM: 142310 Protein ID: AAA61306 [align] AAB59406 [align] CAB06552 [align] RefSeq: NM_005332 [Target %id: 100; Query %id: 100] SWISSPROT: HBAZ_HUMAN [Target %id: 100; Query %id: 99]
GO	The following GO terms have been mapped to this entry via Swissprot/SpTrEMBL: GO:0001524 [globin] GO:0005833 [hemoglobin] GO:0006810 [transport] GO:0015671 [oxygen transport]
InterPro	IPR000971 Globin - [View other Ensembl genes with this domain] IPR002338 Alpha haemoglobin - [View other Ensembl genes with this domain] IPR002339 Pi haemoglobin - [View other Ensembl genes with this domain] IPR002340 Zeta haemoglobin - [View other Ensembl genes with this domain]
Protein Family	ENSF0000000094 : HEMOGLOBIN ALPHA CHAIN This cluster contains 3 Ensembl gene member(s)
Transcript Structure	Exon structure 
Protein Structure	

[top](#)

Transcript-based displays

- Summary
- Supporting evidence (21)
- Sequence
 - Exons (3)
 - cDNA
 - Protein
- External References
 - General identifiers (27)
 - Oligo probes (23)
- Ontology
 - GO graph (17)
 - GO table (17)
- Genetic Variation
 - Variation table
 - Variation image
 - Population comparison
 - Comparison image
- Protein Information
 - Protein summary
 - Domains & features (19)
 - Variations (18)
- External data
 - Personal annotation
- ID History
 - Transcript history
 - Protein history

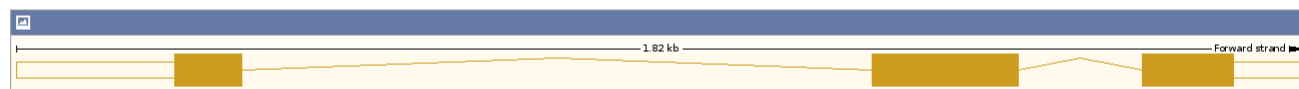
Configure this page
Add your data
Export data
Bookmark this page
Share this page

Transcript: HBZ-001 ENST00000252951

Descriptionhemoglobin, zeta [Source:HGNC Symbol;Acc:HGNC:4835]
SynonymsHBZ-T1, HBZ1
LocationChromosome 16: 152,687-154,503 forward strand.
GeneThis transcript is a product of gene [ENSG00000130656](#)
This gene has 1 transcript (splice variant) [Hide transcript table](#)

Name	Transcript ID	bp	Protein	Biotype	CCDS	RefSeq	Flags
HBZ-001	ENST00000252951	755	142 aa	Protein coding	CCDS10397	NM_005332 NP_005323	TSL:1 GENCODE basic APPRIS P1

Summary



Statistics
CCDSThis transcript is a member of the Human CCDS set: [CCDS10397](#)
UniprotThis transcript corresponds to the following Uniprot identifiers: [P02008](#)
Transcript Support Level1
Ensembl versionENST00000252951.2
TypeKnown protein coding
Prediction MethodTranscript where the Ensembl genebuild transcript and the [Vega](#) manual annotation have the same sequence, for every base pair. See [article](#).
Alternative transcriptsThis transcript corresponds to the following database identifiers:
Havana transcript: [OTTHUMT00000133205](#)
GENCODE basic geneThis transcript is a member of the Gencode basic gene set.

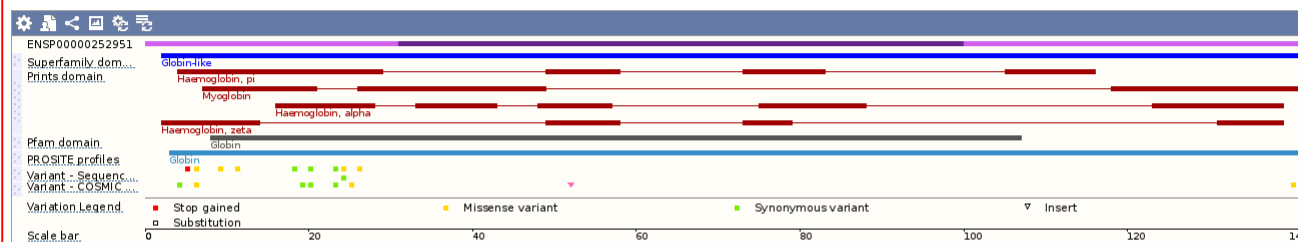
Ensembl release 78 - December 2014 © [WTSI](#) / [EBI](#)

[Permanent link](#) - [View in archive site](#)

[W](#)
[f](#)
[t](#)
[About Ensembl](#)
[Privacy Policy](#)
[Disclaimer](#)
[Contact Us](#)

Protein summary

Protein domains for ENSP00000252951.2



Ensembl release 78 - December 2014 © [WTSI](#) / [EBI](#)

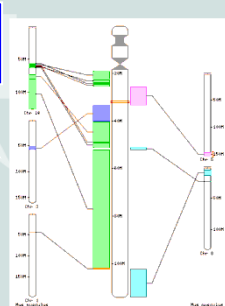
[Permanent link](#) - [View in archive site](#)

[W](#)
[f](#)
[t](#)
[About Ensembl](#)
[Privacy Policy](#)
[Disclaimer](#)
[Contact Us](#)

Genomic alignments

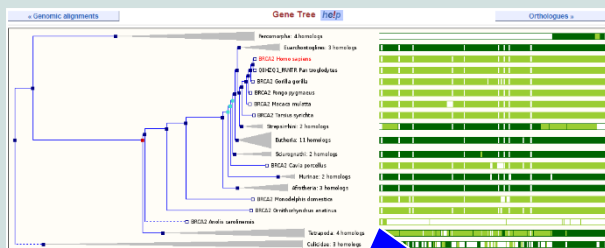
Pick a genome

Synteny



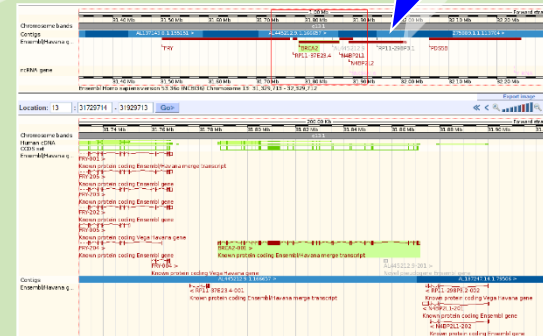
Across species

Orthology



Chromosomes

Genes



Gene families



SNPs

Within species

Biological Data Banks

- Individual sizes anywhere from 1 Mb to 1 Tb
- GenBANK (DNA data bank)
 - 850 Gb (2020)
 - doubling every 18 months
- Total volume of annotated public biological data banks at the moment is approx. 8 Tb
- Disk space required to store *raw* trace data for the original HGP at the Sanger Centre: ~22 Tb
- One NGS experiment can generate 1 Tb of unprocessed raw image data
- Current deposited processed raw NGS data: ~1.4 Pb
- Current Sanger/EBI disk storage: >55 Pb

GenBank Release 122 Statistics (Feb 2001)

80,000 Species

10 million DNA sequences

12 Billion bases, or characters of sequence data

43 Gigabytes of sequence and annotations

GenBank Release 241 Statistics (Dec 2020)

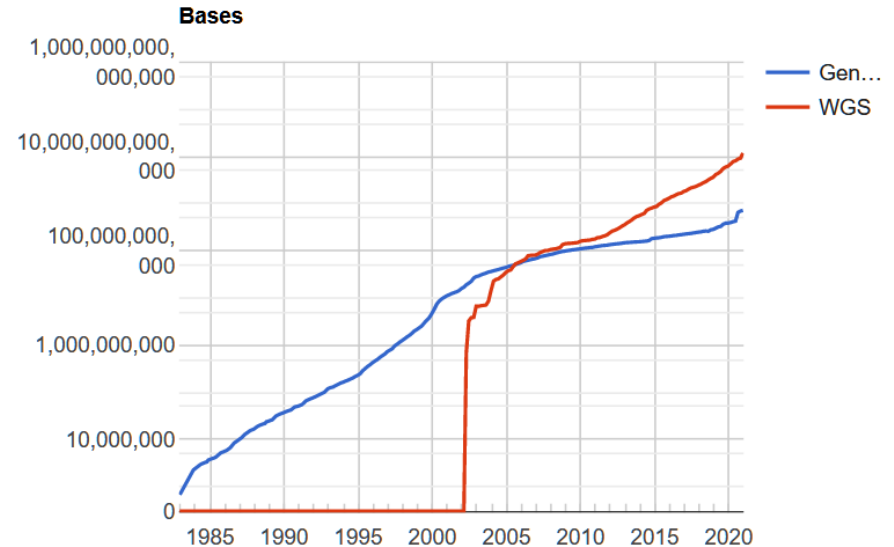
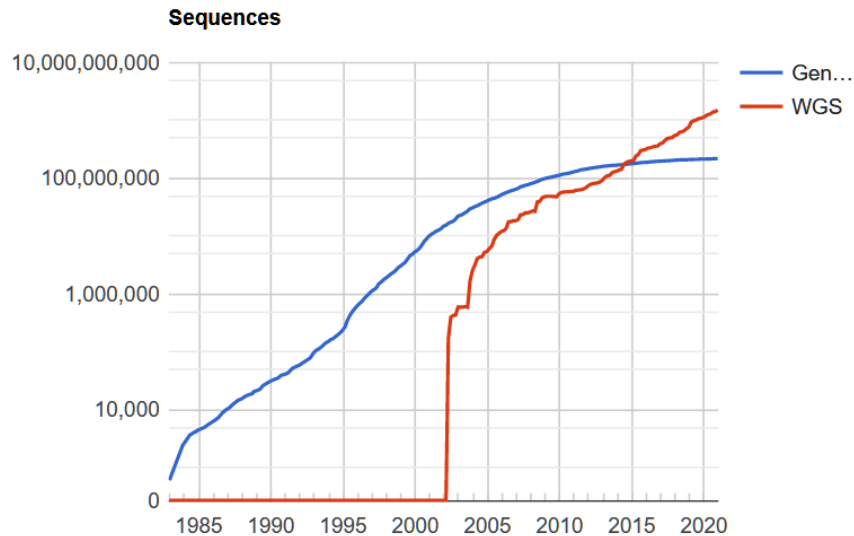
489,144 Species

221 million DNA sequences

723 billion bases, or characters of sequence data

849 Gbytes of sequence and annotations

Total raw sequence data now exceeds total of annotated sequence data



WGS = Whole Genome Shotgun
(raw unfinished sequence data)

Databases & Data Banks

Difference between databases and data banks

A data bank is a collection of data organized in the form of one or more computer files. A database is a collection of organized data along with a program for accessing this data. These terms are frequently (and wrongly) interchanged by biologists.

Most biological databases are in fact technically data banks, but this is rapidly changing.

Types of Data Resource

- Data resources can be characterised by:
 - Type of data (obviously)
 - Data entry and quality control
 - Primary (experimental data) or Secondary (derived data)
 - Technical design
 - Flat file
 - Relational
 - Large Excel file (NO!!)
 - Maintainer status
 - Publicly Core-funded
 - Academic (sustained funding)
 - Commercial
 - Academic (grant funded)
 - Academic (unfunded)
 - Availability
 - Public domain or Creative Commons
 - Commercially restricted academic license
 - Commercial license



RISK

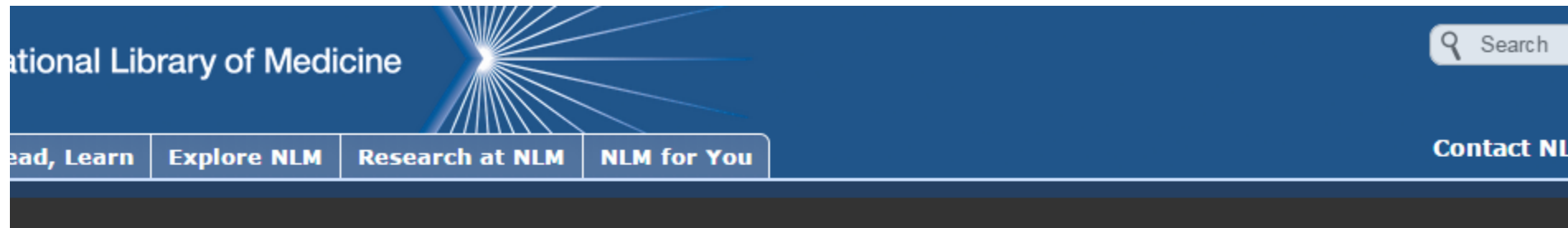
N.B. 50% of Small Medium Enterprises in the UK fail within 5 years!



RISK

Types of Data

Biological Images: The NLM Visible Human Project



The Visible Human Project®

Overview

The Visible Human Project® is an outgrowth of the NLM's 1986 Long-Range Plan. It is the creation of complete, anatomically detailed, three-dimensional representations of the normal male and female human bodies. Acquisition of transverse CT, MR and cryosection images of representative male and female cadavers has been completed. The male was sectioned at one millimeter intervals, the female at one-third of a millimeter intervals.

http://www.nlm.nih.gov/research/visible/visible_human.html



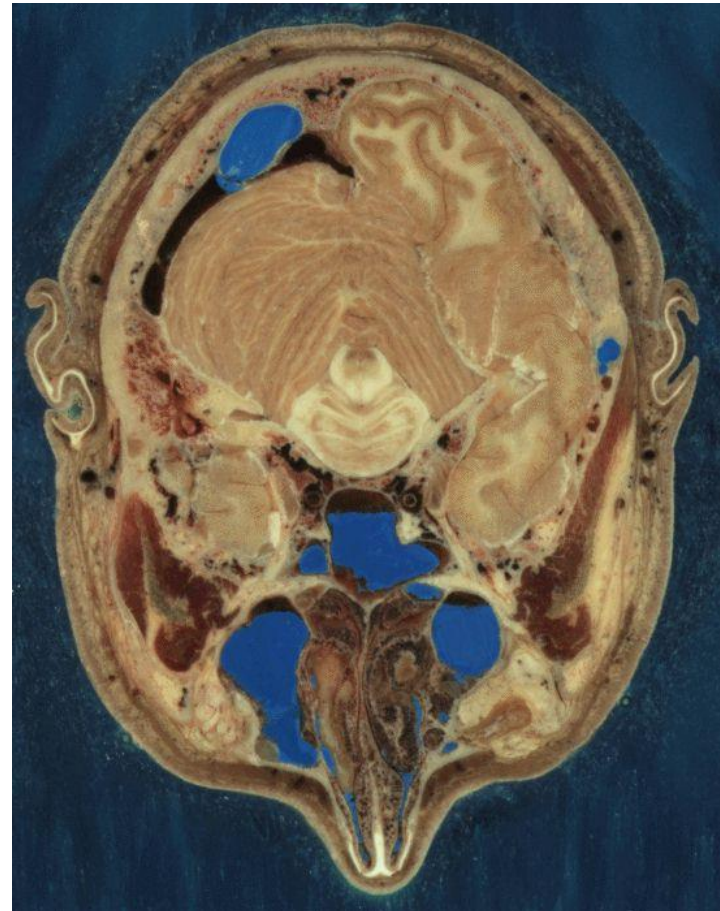
http://www.nlm.nih.gov/images/ad_visibleHumanProject.gif

Biological Images: The NLM Visible Human Project

Long-term Aim:

To create database of voxel data labelled according to e.g. tissue type and expression data.

Section through Visible Human Male - head, including cerebellum, cerebral cortex, brainstem, nasal passages (from Head subset)



Taxonomy Record Fields



The image shows a screenshot of the NCBI Taxonomy Browser interface. At the top, there is a navigation bar with links to PubMed, Entrez, BLAST, OMIM, Taxonomy, and Structure. Below this is a search bar containing the text "Homo sapiens". To the right of the search bar are buttons for "Go" and "Clear", and a "lock" checkbox. Below the search bar, the text "Homo sapiens" is displayed. Underneath, there are several fields: "Taxonomy ID: 9606", "Preferred common name: human", "Rank: species", "Genetic code: [Translation table 1 \(Standard\)](#)", and "Mitochondrial genetic code: [Translation table 2](#)". These two links are highlighted with a red box. Below these fields is a section for "Other names:" which includes "man[common name]". Further down is a section for "Lineage(abbreviated)" which contains a list of taxonomic ranks: "Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Primates; Catarrhini; Hominidae; Homo". This list is also highlighted with a red box. At the bottom, there are radio buttons for "Nucleotide (4898996)", "Protein (128983)", "Structure (2973)", and "Genome records (25)". A "Submit Query" button is located at the bottom left.

NCBI Taxonomy Browser

PubMed Entrez BLAST OMIM Taxonomy Structure

Search for Homo sapiens As complete name lock Go Clear

Homo sapiens

Taxonomy ID: 9606
Preferred common name: human
Rank: species
Genetic code: [Translation table 1 \(Standard\)](#)
Mitochondrial genetic code: [Translation table 2](#)

Other names:
man[common name]

Lineage(abbreviated):
[Eukaryota](#); [Metazoa](#); [Chordata](#); [Craniata](#); [Vertebrata](#); [Euteleostomi](#); [Mammalia](#); [Eutheria](#); [Primates](#); [Catarrhini](#); [Hominidae](#); [Homo](#)

☒ Nucleotide (4898996) ☐ Protein (128983) ☐ Structure (2973) ☐ Genome records (25)

Submit Query

Associated
Genetic Codes

Lineage

Ontological Data

- Gene Ontology database – GO
 - Maintained in mySQL (RDBMS)
 - Available as XML
- Other ontologies being actively developed
 - Experiment ontology
 - Will allow arbitrary biological experiments to be described in a systematic way
 - E.g. temperature, reagents, cell lines, organisms etc.

Ontological Data

EMBL-EBI

A fast browser for Gene Ontology terms and annotations.

- QuickGO
 - Help
 - Reference
 - FAQs
 - Video tutorials
 - Downloads
 - geneontology.org
 - UniProt-GOA project
 - Web Services

EBI > Databases > QuickGO

QuickGO



 Web Services  Dataset  Term Basket: 0 

Search and Filter GO annotation sets 

Extensive filters are available from this page to allow the generation of specific subsets of GO annotations, mapped to sequence identifiers of your choice.

Investigate GO slims 

GO slims are lists of GO terms that have been selected from the full set of terms available from the Gene Ontology project.

GO slims can be used to generate a focused view of part of the GO, or with annotation data they can be used to see how a set of proteins/genes can be broadly categorized (using annotation data and the relationships that exist between terms in the ontologies).

Further information on GO slims can be found at the [GO Consortium web site](http://www.geneontology.org).

Ontological Data



A fast browser for Gene Ontology terms and annotations.

EBI > Databases > QuickGO

GO:0008152 metabolic process



 Web Services  Dataset  Term Basket: 0 

Term Information

Ancestor Chart

Child Terms

Protein Annotation

Co-occurring Terms

Change Log

This table lists all terms that are direct descendants (child terms) of GO:0008152:

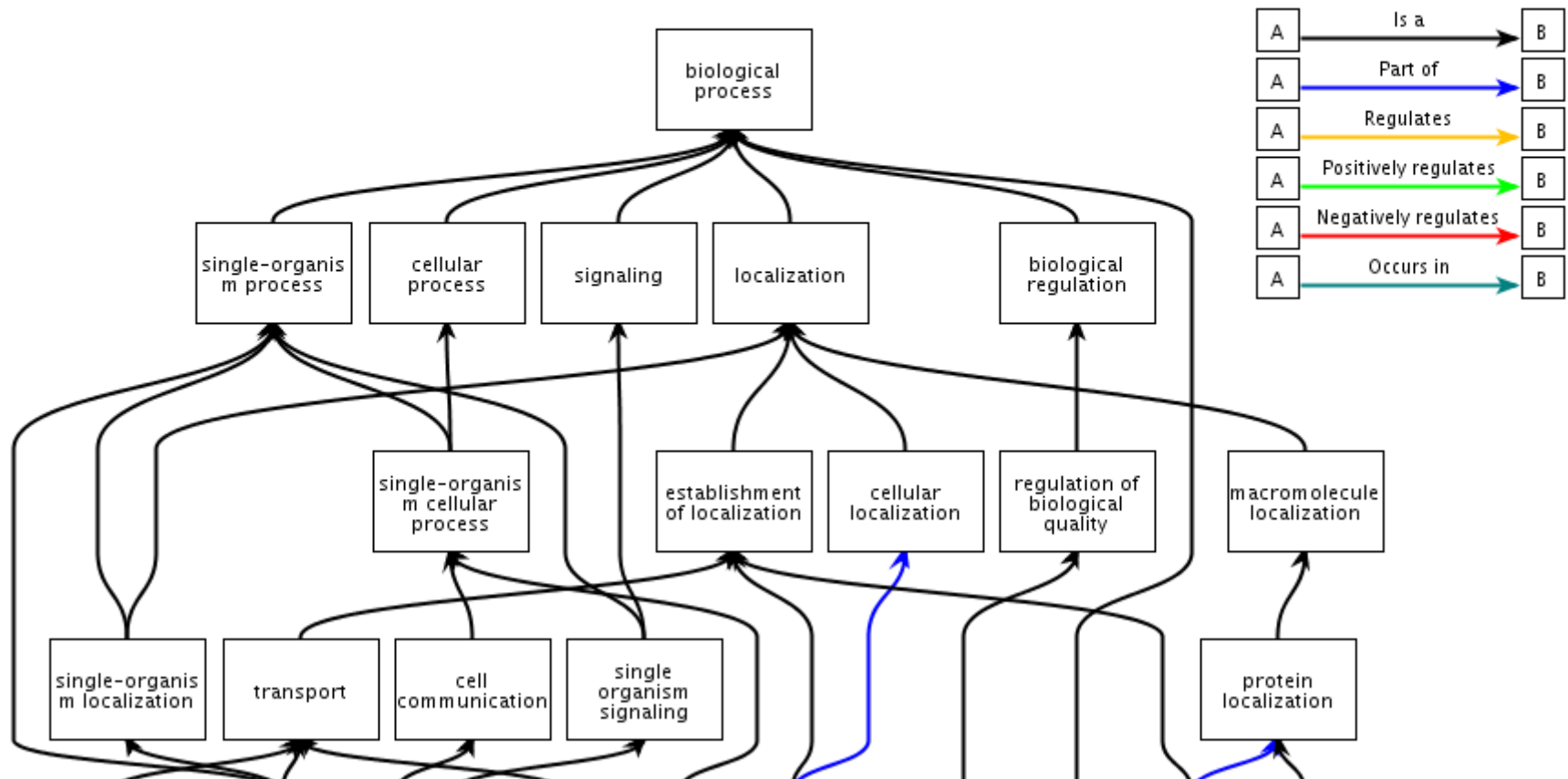
Relationship To GO:0008152	Child Term	Child Term Name
Negatively regulates	 GO:0009892	negative regulation of metabolic process
Positively regulates	 GO:0009893	positive regulation of metabolic process
Regulates	 GO:0019222	regulation of metabolic process
Is a	 GO:0009056	catabolic process
Is a	 GO:0009058	biosynthetic process
Is a	 GO:0044237	cellular metabolic process
Is a	 GO:0044238	primary metabolic process
Is a	 GO:0044033	multi-organism metabolic process
Is a	 GO:0006807	nitrogen compound metabolic process
Part of	 GO:0003824	catalytic activity

Ontological Data

     Web Services Dataset Term Basket: 0

[Term Information](#) [Ancestor Chart](#) [Child Terms](#) [Protein Annotation](#) [Co-occurring Terms](#) [Change Log](#)

This chart is interactive; you can click on the term boxes and legend for more information.



Molecular Sequence Data

- The major task in computational molecular biology is currently to “decipher” information contained in biological sequences
- Since the nucleotide sequence of a genome contains all information necessary to produce a functional organism, we should in theory be able to duplicate this decoding using computers

Structure

- Macromolecular structure divided into
 - **primary** structure (1D sequence)
 - **secondary** structure (local 2D & 3D)
 - **tertiary** structure (global 3D)
- DNA composed of four **nucleotides** or "bases":
A,C,G,T
- RNA composed of four also: A,C,G,U (T transcribed as U)
- Proteins are composed of **amino acids**

Sequence Features

- A **sequence** is a linear set of characters (sequence elements) representing nucleotides or amino acids
- A **sequence feature** is a pattern that is observed to occur in more than one sequence and (usually) to be correlated with some function

Sequence Features

- Features following an exact pattern
 - restriction enzyme recognition sites
- Features with approximate patterns
 - promoters
 - transcription initiation sites
 - transcription termination sites
 - polyadenylation sites
 - ribosome binding sites
 - protein features

Character Representation of Sequences

- DNA or RNA
 - use 1-letter codes (e.g., A,C,G,T)
- Protein
 - use 1-letter codes
 - can convert to/from 3-letter codes

The I.U.B. Nucleic Acid Code

- A, C, G, T, U
- R = A, G (puRine)
- Y = C, T (pYrimidine)
- S = G, C (SStrong hydrogen bonds)
- W = A, T (WWeak hydrogen bonds)
- M = A, C (aMino group)
- K = G, T (Keto group)
- B = C, G, T (not A)
- D = A, G, T (not C)
- H = A, C, T (not G)
- V = A, C, G (not T/U)
- N = A, C, G, T/U (iNdeterminate) X or - are sometimes used

Examples of ASCII sequence file formats

- Fasta
 - Minimal sequence file format
 - Widely used as input file format

```
>gi|995614|dbj|D49653|RATOBESSE Rat mRNA for obese.  
CCAAGAAGAAGAAGACCCCAGCGAGGAAAATGTGCTGGAGACCCCTGTGCCGGTTCCTGTGGCTTTGGTC  
CTATCTGTCCTATGTTCAAGCTGTGCCTATCCACAAAGTCCAGGATGACACCAAACCCCTCATCAAGACC  
ATTGTCACCAGGATCAATGACATTTACACACGCAGTCGGTATCCGCCAGGCAGAGGGTCACCGGTTTGG  
ACTTCATTCCCGGGCTTCACCCCATTTCTGAGTTTGTCCAAGATGGACCAGACCCTGGCAGTCTATCAACA  
GATCCTCACCAGCTTGCCTTCCCAAACGTGCTGCAGATAGCTCATGACCTGGAGAACCTGCGAGACCTC  
CTCCATCTGCTGGCCTTCTCCAAGAGCTGCTCCCTGCCGCAGACCCGTGGCCTGCAGAAGCCAGAGAGCC  
TGGATGGCGTCCTGGAAGCCTCGCTCTACTCCACAGAGGTGGTGGCTCTGAGCAGGCTGCAGGGCTCTCT  
GCAGGACATTCTTCAACAGTTGGACCTTAGCCCTGAATGCTGAGGTTTC
```

An important derivative of FASTA format is the FASTQ format for sequence *reads*

- FASTQ

- Gives both the sequence AND confidence of the base call as given by the sequencing machine's post-processing software
- Quality is indicated by ASCII character set:
`!"#$%&'()*+,-./0123456789:;<=>?@ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_`abcdefghijklmnopqrstuvwxyz{|}~`
- Widely used as input file format for gene expression assembly from RNAseq data

```
@SRR001666.1 071112_SLXA-EAS1_s_7:5:1:817:345 length=36
GGGTGATGGCCGCTGCCGATGGCGTCAAATCCCACC
+SRR001666.1 071112_SLXA-EAS1_s_7:5:1:817:345 length=36
IIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIIII9IG9IC
```

Only the bold parts are universally required.

Both '@' and '+' can also appear in the quality string! Perhaps we shouldn't leave file format design to biologists...

Examples of ASCII sequence file formats

GenBank

```
LOCUS      RATOBESE      539 bp ss-mRNA      ROD      23-SEP-1995
DEFINITION Rat mRNA for obese.
ACCESSION  D49653
KEYWORDS   .
SOURCE     Rattus norvegicus (strain OLETF, LETO and Zucker, ) differentiated
           adipose cDNA to mRNA.
  ORGANISM Rattus norvegicus
           Eukaryotae; mitochondrial eukaryotes; Metazoa; Chordata;
           Vertebrata; Sarcopterygii; Mammalia; Eutheria; Rodentia;
           Sciurognathi; Myomorpha; Muridae; Murinae; Rattus.
REFERENCE  1 (bases 1 to 539)
  AUTHORS  Murakami,T. and Shima,K.
  TITLE    Cloning of rat obese cDNA and its expression in obese rats
  JOURNAL  Biochem. Biophys. Res. Commun. 209, 944-952 (1995)
  STANDARD full automatic
COMMENT    Submitted (10-Mar-1995) to DDBJ by:
           Takashi Murakami
           Department of Laboratory Medicine
           School of Medicine
           University of Tokushima
           Kuramotocho 3-chome
           Tokushima 770
           Japan
           Phone: +81-886-33-7184
           Fax:   +81-886-31-9495.
```

[continued]

Examples of ASCII sequence file formats

GenBank [continued]

```
NCBI gi: 995614
FEATURES
    source
        Location/Qualifiers
            1..539
                /organism="Rattus norvegicus"
                /strain="OLETF, LETO and Zucker"
                /dev_stage="differentiated"
                /sequenced_mol="cDNA to mRNA"
                /tissue_type="adipose"
    CDS
        30..533-
            /partial
            /note="NCBI gi: 995615"
            /codon_start=1
            /product="obese"
            /translation="MCWRPLCRFLWLWSYLSYVQAVPIHKVQDDTKTLIKTIVTRIND
            ISHTQSVSARQRVGTGLDFIPGLHPILSLSKMDQTLAVYQQILTSLPSONVLQIAHDLE
            NLRDLLHLLAFSKSCSLPQTRGLQKPESLDGVLEASLYSTEVVALSRLQGSLLQDILQQ
            LDLSPEC"
BASE COUNT      121 a      167 c      133 g      118 t
ORIGIN
    1 ccaagaagaa gaagacccca gcgaggaaaa tgtgctggag acccctgtgc cggttcctgt
    61 ggctttggtc ctatctgtcc tatgttcaag ctgtgcctat ccacaaagtc caggatgaca
    121 ccaaaaccct catcaagacc attgtcacca ggatcaatga catttcacac acgcagtcgg
    181 tatccgccag gcagagggtc accggtttgg acttcattcc cgggcttcac cccattctga
    241 gtttggtccaa gatggaccag accctggcag tctatcaaca gatcctcacc agcttgcctt
    301 cccaaaacgt gctgcagata gctcatgacc tggagaacct gcgagacctc ctccatctgc
    361 tggccttctc caagagctgc tccctgccgc agaccctgtg cctgcagaag ccagagagcc
    421 tggatggcgt cctggaagcc tcgctctact ccacagaggt ggtggctctg agcaggctgc
    481 agggctctct gcaggacatt cttcaacagt tggaccttag ccctgaatgc tgaggtttc
//
```

The ENA (European Nucleotide Archive)

- The ENA is a high level repository for different nucleotide-related data sets
- The archive is composed of three main databases: the Sequence Read Archive, the Trace Archive and the EMBL Nucleotide Sequence Database (also known as EMBL-bank)
- The EMBL Nucleotide Sequence Database component is the equivalent of Genbank
- NCBI maintains Genbank separately from its read and trace archives

◉ Please subscribe to ena-announce mailing list here: listserv.ebi.ac.uk/mailman/listin... to receive alerts about ENA services.

Sequence: V00491.1

Human gene for alpha 1 globin.

[Send Feedback](#)

View: [TEXT](#) [FASTA](#) [XML](#)

Download: [XML](#) [FASTA](#) [TEXT](#)

Organism Homo sapiens	Molecule type genomic DNA	Topology linear	Data class STD	Taxonomic Division HUM
Sequence length 900	Sequence Version 1	First public 09-JUN-1982	Last updated 14-NOV-2006	Show Version History V00491

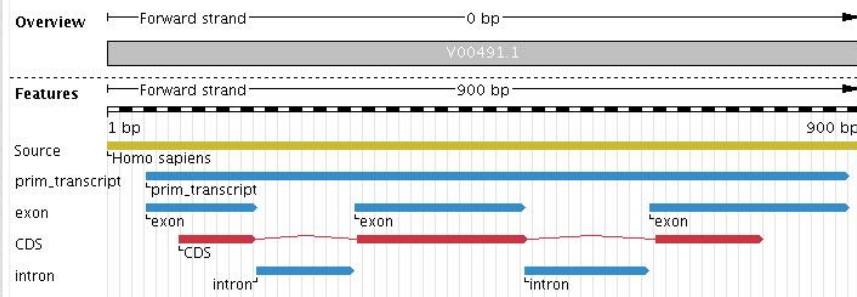
Keywords
alpha-globin, germ line, globin.

Lineage

[Eukaryota](#), [Metazoa](#), [Chordata](#), [Craniata](#), [Vertebrata](#), [Euteleostomi](#), [Mammalia](#), [Eutheria](#), [Euarchontoglires](#), [Primates](#), [Haplorrhini](#), [Catarrhini](#), [Hominidae](#), [Homo](#)

[Navigation](#) [Overview](#) [Source Feature\(s\)](#) [Comments](#) [Sequence](#) [Publications](#) [Other Feature\(s\)](#)

Base range: - [Apply](#)



EMBL Data Bank format is similar (but annoyingly different) to Genbank

```
ID V00491; SV 1; linear; genomic DNA; STD; HUM; 900 BP.
XX
AC V00491;
XX
DT 09-JUN-1982 (Rel. 01, Created)
DT 14-NOV-2006 (Rel. 89, Last updated, Version 7)
XX
DE Human gene for alpha 1 globin.
XX
KW alpha-globin; germ line; globin.
XX
OS Homo sapiens (human)
OC Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia;
OC Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae;
OC Homo.
XX
RN [1]
RP 1-900
RX DOI; 10.1016/0092-8674(80)90347-5.
RX PUBMED; 7448866.
RA Michelson A.M., Orkin S.H.;
RT "The 3' untranslated regions of the duplicated human alpha-globin genes are
RT unexpectedly divergent";
RL Cell 22(2 Pt 2):371-377(1980).
XX
DR MD5; 6f21680c81c0f8e98a60926bb73bdd70.
DR EPD; EP07071; HS_HBA1.
DR EPD; EP53001; HS_HBA2.
DR Ensembl-Gn; ENSG00000188536; homo_sapiens.
DR Ensembl-Gn; ENSG00000206172; homo_sapiens.
DR Ensembl-Tr; ENST00000251595; homo_sapiens.
DR Ensembl-Tr; ENST00000320868; homo_sapiens.
DR EuropePMC; PMC2529266; 18657265.
XX
CC KST HSA.ALP1GLOBIN.GL [900]
XX
FH Key Location/Qualifiers
FH
FT source 1..900
FT /organism="Homo sapiens"
FT /mol_type="genomic DNA"
```

```

FT          /db_xref="taxon:9606"
FT prim_transcript 47..888
FT exon          47..179
FT          /number=1
FT CDS          join(87..179,297..500,650..778)
FT          /product="alpha 1 globin"
FT          /db_xref="GOA:P69905"
FT          /db_xref="UniProtKB/Swiss-Prot:P69905"
FT          /protein_id="CAA23750.1"
FT          /translation="VLSPADKTNVKAAWGKVGAGHAGEYGAEALERMFLSFPTTKTYFPH
FT          FDLSHGSAQVKGHGKKVADALTNAVAHVDDMPNALSALSDLHAHKLRVDPVNFKLLSHC
FT          LLVTLAAHLPAEFTPAVHASLDKFLASVSTVLTSKYR"
FT intron        180..296
FT          /number=1
FT exon          297..500
FT          /number=2
FT intron        501..649
FT          /number=2
FT exon          650..888
FT          /number=3

```

XX

```

SQ Sequence 900 BP; 139 A; 349 C; 260 G; 152 T; 0 other;
tgcccccgcg cccaagcat aaacctggc gcgtcgcg cccggcactc ttctggtccc      60
cacagactca gagagaaccc accatggtgc tgtctcctgc cgacaagacc aacgtcaagg      120
ccgcctgggg taaggtcggc gcgcacgctg gcgagtatgg tgcggaggcc ctggagaggt      180
gaggtccct cccctgctcc gacccgggct cctcgccgcg cccgaccac aggccaccct      240
caaccgtcct ggccccggac ccaaacccca cccctcactc tgcttctccc cgcaggatgt      300
tcctgtcctt cccaaccacc aagacctact tccgcactt cgacctgagc cacggctctg      360
cccaggttaa ggccacggc aagaagggtg ccgacgcgct gaccaacgcc gtggcgcacg      420
tggaacgacat gcccaacgcg ctgtccgccc tgagcgacct gcacgcgcac aagcttcggg      480
tggaacgggt caacttcaag gtgagcgggc ggccgggagc gatctgggtc gaggggagcg      540
atggcgccct cctcgccagg cagaggatca cgcgggttgc gggagggtgta gcgcaggcgg      600
cggctgcgga cctgggccct cgccccact gacctcttct tctgcacagc tcctaagcca      660
ctgacctgtg gtgacctgg ccgccacact cccgcgcgag ttcacccctg cggtgacgcg      720
ctccctggac aagtctcctg cttctgtgag caccgtgctg acctccaaat accgttaagc      780
tggaacgctg gtggccatgc ttcttgcccc ttgggcctcc cccagcccc tcctccctt      840
cctgcacccg taccctcggt gtctttgaat aaagtctgag tgggcggcag cctgtgtgtg      900

```

//

Sequence variant data e.g. 1000 Genomes Project Data (<http://www.1000genomes.org>)

The most commonly
observed variants
are SNPs (Single
Nucleotide
Polymorphisms). →

```

          97490      97500      97510      975
+-----+-----+-----+-----+
GCGTCTCGTATCATATTTTTCAGGACATCATCTATCC
:gcggtctcgtatcatatTTTTT
:gcggtctcgtatcatatTTTTT
:gcggtctcgtatcatatgTTTTTca
:gcggtctcgtatcatatTTTTTcag
:gcggtctcgtatcatatgTTTTTcagg
:gcggtctcgtatcatatTTTTTcaggga
:gcggtctcgtatcatatTTTTTcaggacat
:gcggtctcgtatcatatTTTTTcaggaca
:gcggtctcgtatcatatTTTTTcaggaca
:gcggtctcgtatcatatTTTTTcaggaca
:gcggtctcgtatcatatgTTTTTcaggacatca
          gtatcatatTTTTTcaggacatcatctatcg
  
```

Careful statistical analysis
of assembly data is
needed to distinguish
between population
variants and sequencing
errors! SNPs are typically
defined when variant is
observed in > 1% of
population.

The most widely used format for storing and retrieving such
variant data is the Variant Call Format (VCF)...

TAB separated records...

#CHROM	POS	ID	REF	ALT	QUAL	FILTER	INFO	FORMAT	NA00001	NA00002
20	14370	rs6054257	G	A	29	0	NS=3;DP=14;AF=0.5;DB;H2	GT:GQ:DP:HQ	0 0:48:1:51,51	1 0:48:8:51,51
20	17330	.	T	A	3	q10	NS=3;DP=11;AF=0.017	GT:GQ:DP:HQ	0 0:49:3:58,50	0 1:3:5:65,3
20	1110696	rs6040355	A	G,T	67	0	NS=2;DP=10;AF=0.333,0.667;AA=T;DB	GT:GQ:DP:HQ	1 2:21:6:23,27	2 1:2:0:18,2
20	1230237	.	T	.	47	0	NS=3;DP=13;AA=T	GT:GQ:DP:HQ	0 0:54:7:56,60	0 0:48:4:51,51

Each sample (patient) ...
('|' indicated this is phased
data, '/' if not)

Examples of ASCII sequence file formats

Swiss-Prot / UniProt

```
ID      GLBH TRICO          STANDARD;          PRT;    173 AA.
AC      P276T3;
DT      01-AUG-1992 (Rel. 23, Created)
DT      01-AUG-1992 (Rel. 23, Last sequence update)
DT      01-JUN-1994 (Rel. 29, Last annotation update)
DE      GLOBIN-LIKE HOST-PROTECTIVE ANTIGEN PRECURSOR.
OS      Trichostrongylus colubriformis.
OC      Eukaryota; Metazoa; Nematoda; Chromadorea; Rhabditida; Strongylida;
OC      Trichostrongyloidea; Trichostrongylidae; Trichostrongylinae;
OC      Trichostrongylus.
OX      NCBI_TaxID=6319;
RN      [1] -
RP      SEQUENCE FROM N.A., AND SEQUENCE OF 16-37 AND 163-173.
RC      STRAIN=MCMaster;
RX      MEDLINE=92178288; PubMed=1542314;
RA      Frenkel M.J., Dopheide T.A.A., Wagland B.M., Ward C.W.;
RT      "The isolation, characterization and cloning of a globin-like, host-
RT      protective antigen from the excretory-secretory products of
RT      Trichostrongylus colubriformis.";
RL      Mol. Biochem. Parasitol. 50:27-36(1992).
CC      -!- FUNCTION: MAY BE A GLOBIN AND MAY PLAY A ROLE IN OXYGEN TRANSPORT.
CC      -!- SUBCELLULAR LOCATION: EXTRACELLULAR.
DR      EMBL; M63263; AAA30102.1; -.
DR      PIR; S29131; S29131.
DR      HSSP; P28316; 1ASH.
DR      InterPro; IPR000971; -.
DR      Pfam; PF00042; globin; 1.
DR      PROSITE; PS01033; GLOBIN; 1.
KW      Heme; Oxygen transport; Respiratory protein; Signal; Antigen.
```

Examples of ASCII sequence file formats

Swiss-Prot / UniProt (cont.)

```
FT      SIGNAL           1       15
FT      CHAIN            16      173      GLOBIN-LIKE HOST-PROTECTIVE ANTIGEN.
FT      METAL            114     114      IRON (HEME) (BY SIMILARITY) .
FT      VARIANT          126     126      E -> D (IN ADES3/2 CLONE) .
FT      VARIANT          129     129      S -> G (IN ADES3/2 CLONE) .
SQ      SEQUENCE        173 AA;  19988 MW;  74019C76C18BE6ED CRC64;
MRFLLLAAFV AYAYAKSDEE IRKDALSALD VVPLGSTPEK LENGREFYKY FFTNHQDLRK
YFKGAETFTA DDIAKSDRFK KLGNQLLLSV HLAADTYDNE MIFRAFVRDT IDRHVDRGLD
PKLWKEFWSI YQKFLESKGK TLSADQKAAF DAIGTRFNDE AQKQLAHHGL PHT
//
```


Examples of ASCII sequence file formats

PDB (Protein Data Bank)

HEADER	OXYGEN TRANSPORT	07-MAR-84	4HHB	4HHB	3
COMPND	HEMOGLOBIN (DEOXY)			4HHB	4
SOURCE	HUMAN (HOMO SAPIENS)			4HHB	5
AUTHOR	G.FERMI,M.F.PERUTZ			4HHB	6
REVDAT	2 15-OCT-89 4HHBA 3	MTRIX		4HHBA	1
REVDAT	1 17-JUL-84 4HHB 0			4HHB	7
SPRSDE	17-JUL-84 4HHB 1HHB			4HHB	8
JRNL	AUTH G.FERMI,M.F.PERUTZ,B.SHAANAN,R.FOURME			4HHB	9
JRNL	TITL THE CRYSTAL STRUCTURE OF HUMAN DEOXYHAEMOGLOBIN AT			4HHB	10
JRNL	TITL 2 1.74 ANGSTROMS RESOLUTION			4HHB	11
JRNL	REF J.MOL.BIOL.	V. 175 159 1984		4HHB	12
JRNL	REFN ASTM JMOBAK UK ISSN 0022-2836		070	4HHB	13

Examples of ASCII sequence file formats

PDB (cont.)

```
REMARK      3                                                    4HHB  71
REMARK      3 REFINEMENT. UNRESTRAINED REFINEMENT.  THE CONFORMATION OF  4HHB  72
REMARK      3 THE HEME GROUP WAS MODIFIED BEFORE STARTING THE  4HHB  73
REMARK      3 UNRESTRAINED REFINEMENT.  THE FINAL R VALUE IS 0.135.  4HHB  74
REMARK      4                                                    4HHB  75
REMARK      4 THE CRYSTALLOGRAPHIC ASYMMETRIC UNIT CONTAINS TWO ALPHA AND  4HHB  76
REMARK      4 TWO BETA CHAINS. ONLY ONE CHAIN OF EACH TYPE IS REPRESENTED  4HHB  77
REMARK      4 HERE.                                                    4HHB  78
REMARK      5
...
REMARK      5 XYZ coordinates for atoms in
                  Angstroms (1 A = 0.1 nm)
...                                                    4HHB  79
ATOM         1  N    VAL A    1    6.204  16.869  4.854  7.00  49.05    4HHB  205
ATOM         2  CA   VAL A    1    6.913  17.759  4.607  6.00  43.14    4HHB  206
ATOM         3  C    VAL A    1    8.504  17.378  4.797  6.00  24.80    4HHB  207
ATOM         4  O    VAL A    1    8.805  17.011  5.943  8.00  37.68    4HHB  208
ATOM         5  CB   VAL A    1    6.369  19.044  5.810  6.00  72.12
```

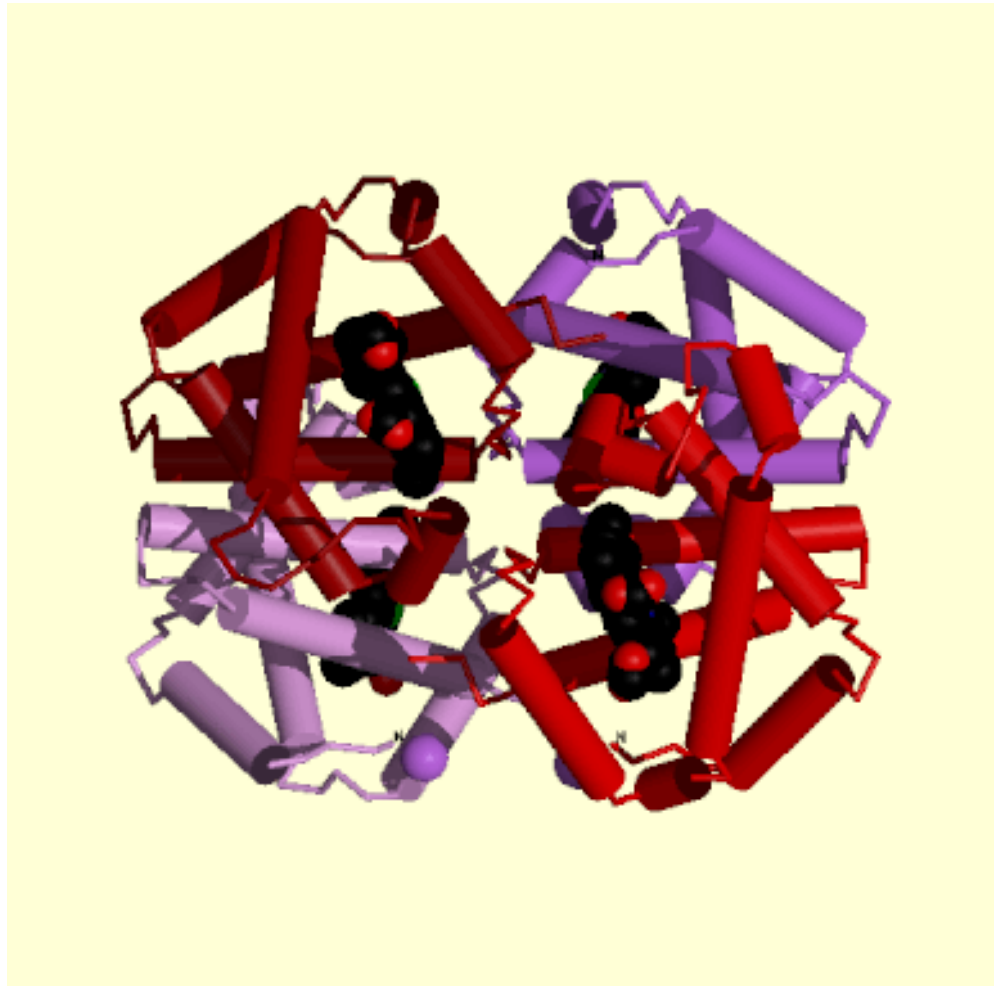
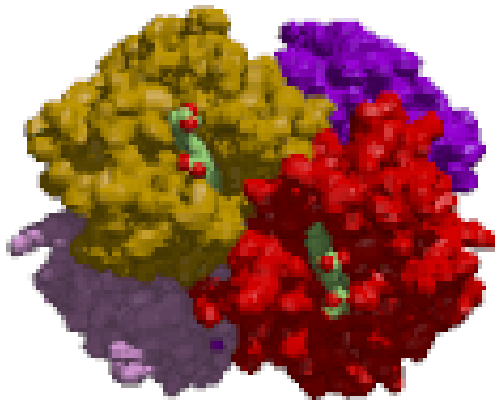
PDB format is now deprecated in favour of the more flexible mmCIF format

```
loop_
_atom_site.group_PDB
_atom_site.id
_atom_site.type_symbol
_atom_site.label_atom_id
_atom_site.label_alt_id
_atom_site.label_comp_id
_atom_site.label_asym_id
_atom_site.label_entity_id
_atom_site.label_seq_id
_atom_site.pdbx_PDB_ins_code
_atom_site.Cartn_x
_atom_site.Cartn_y
_atom_site.Cartn_z
_atom_site.occupancy
_atom_site.B_iso_or_equiv
_atom_site.pdbx_formal_charge
_atom_site.auth_seq_id
_atom_site.auth_comp_id
_atom_site.auth_asym_id
_atom_site.auth_atom_id
_atom_site.pdbx_PDB_model_num
```

```
ATOM 1 N N . VAL A 1 1 ? 6.204 16.869 4.854 1.00 49.05 ? 1 VAL A N 1
ATOM 2 C CA . VAL A 1 1 ? 6.913 17.759 4.607 1.00 43.14 ? 1 VAL A CA 1
ATOM 3 C C . VAL A 1 1 ? 8.504 17.378 4.797 1.00 24.80 ? 1 VAL A C 1
ATOM 4 O O . VAL A 1 1 ? 8.805 17.011 5.943 1.00 37.68 ? 1 VAL A O 1
ATOM 5 C CB . VAL A 1 1 ? 6.369 19.044 5.810 1.00 72.12 ? 1 VAL A CB 1
ATOM 6 C CG1 . VAL A 1 1 ? 7.009 20.127 5.418 1.00 61.79 ? 1 VAL A CG1 1
ATOM 7 C CG2 . VAL A 1 1 ? 5.246 18.533 5.681 1.00 80.12 ? 1 VAL A CG2 1
ATOM 8 N N . LEU A 1 2 ? 9.096 18.040 3.857 1.00 26.44 ? 2 LEU A N 1
ATOM 9 C CA . LEU A 1 2 ? 10.600 17.889 4.283 1.00 26.32 ? 2 LEU A CA 1
ATOM 10 C C . LEU A 1 2 ? 11.265 19.184 5.297 1.00 32.96 ? 2 LEU A C 1
ATOM 11 O O . LEU A 1 2 ? 10.813 20.177 4.647 1.00 31.90 ? 2 LEU A O 1
ATOM 12 C CB . LEU A 1 2 ? 11.099 18.007 2.815 1.00 29.23 ? 2 LEU A CB 1
ATOM 13 C CG . LEU A 1 2 ? 11.322 16.956 1.934 1.00 37.71 ? 2 LEU A CG 1
ATOM 14 C CD1 . LEU A 1 2 ? 11.468 15.596 2.337 1.00 39.10 ? 2 LEU A CD1 1
ATOM 15 C CD2 . LEU A 1 2 ? 11.423 17.268 0.300 1.00 37.47 ? 2 LEU A CD2 1
ATOM 16 N N . SER A 1 3 ? 11.584 18.730 6.148 1.00 28.01 ? 3 SER A N 1
ATOM 17 C CA . SER A 1 3 ? 12.263 19.871 7.087 1.00 26.03 ? 3 SER A CA 1
ATOM 18 C C . SER A 1 3 ? 13.304 20.329 6.300 1.00 25.99 ? 3 SER A C 1
```

**Classic PDB format
will live on for many
years thanks to
legacy software!**

PDB Entry 4HHB



Secondary Databases

- Contain data which is derived from primary data, sometimes by manual curation, but also by applying algorithms
- Examples:
 - InterPro: collection of patterns/fingerprints for protein families
 - CATH: hierarchical classification of protein structures
 - Pfam: collection of protein families
 - OMIM: information on inherited disease
- NOTE: Swiss-Prot and Uniprot *are* considered *primary* databases, because they archive experimental data. However, Swiss-Prot is *manually curated* and *both* databases contain information that is derived from purely *computational analysis*!
 - Sadly, the world isn't perfect (certainly not in biology!)

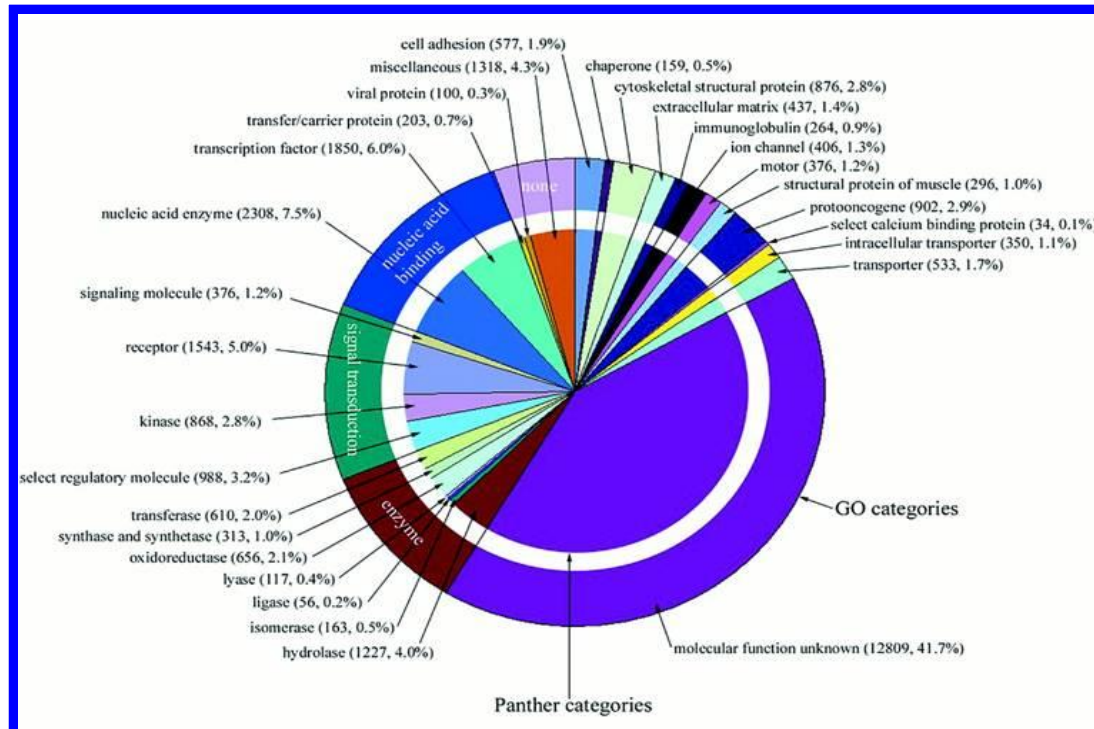
CASE STUDY

Prediction of Protein Function from Sequence Derived Protein Features

REF: Jensen et al. J. Mol. Biol., 319:1257-1265, 2002

(see Moodle)

Many proteins encoded in the Human Genome have unknown Function



How can we infer function using computational methods?

- Directly from properties/features of the sequence
 - *ab initio*
- By homology i.e. using evolution
 - Related proteins often have similar functions
- From the structure of the protein
 - The structure of a protein gives its function
- From microarray data or protein-protein interaction networks

Sequence Alignment

- > 30% Identity between two protein sequences implies probable common structure and possibly common function
- However, there are many exceptions to this “rule of thumb”

```

                10          20          30          40          50
1PLC    --IDVLLGADDGSLAFVPSEFSISPGEKIVFKNNAGFPHNIVFEDEDSIPS-GVDASKISM
        : X:.: : :: :...::... . : ::   :::::      :. ... :. :.
1NIN    ETYTVKLGSDDKGLLVFEPAKLTIKPGDTVEFLNNKVPPHNVVFDAALNPASADLAK-SL
                10          20          30          40          50

                60          70          80          90
1PLC    SEEDLLNAKGETFEVAL---SNKGESFYCSPHQGAGMVGKVTVN-
        :...:X . ...   .   ::::: :.....:
1NIN    SHKQLLMSPGQSTSTTFPADAPAGEYTIFYCEPHRGAGMVGKITVAG
                60          70          80          90         100

```

Transferring Function by Homology

Enzyme Nomenclature

EC 1 Oxidoreductases

EC 1.1

EC 1.1 Acting on the CH-OH group of donors

EC 1.1.1 With NAD⁺ or NADP⁺ as acceptor

EC 1.1.1.1 alcohol dehydrogenase

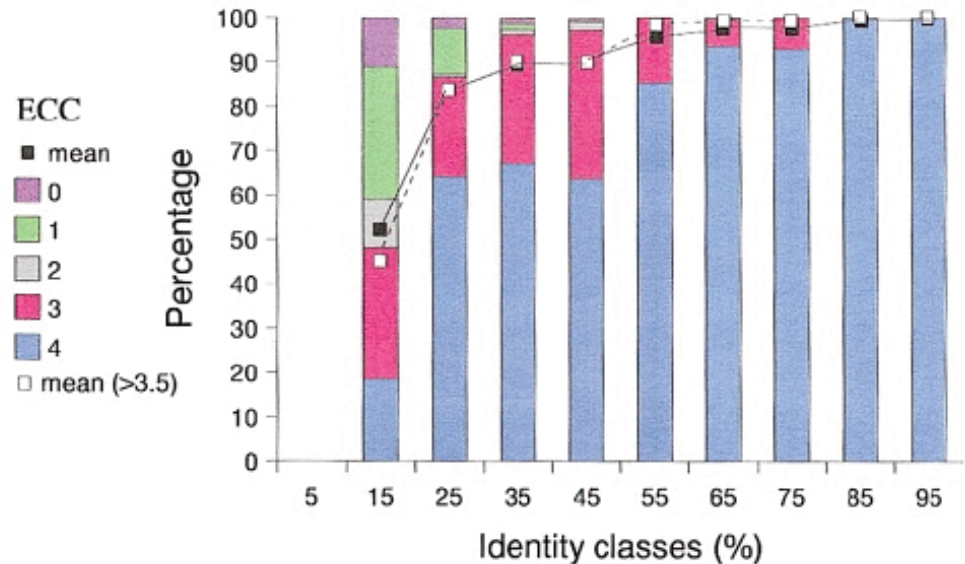
EC 1.1.1.2 alcohol dehydrogenase (NADP⁺)

EC 1.1.1.3 homoserine dehydrogenase

EC 1.1.1.4 (*R,R*)-butanediol dehydrogenase

EC 1.1.1.5 acetoin dehydrogenase

Enzymatic function comparison



Main Steps in applying Intelligent Systems to Bioinformatics Problems

- Identify and extract training data
 - Split into TRAINING and TEST sets
- Identify appropriate FEATURE SET and suitable ENCODING SCHEME
- TRAIN on known examples
- TEST on OTHER known examples (CROSS-VALIDATION)
- Evaluate performance
 - Take into account the required application e.g. FALSE NEGATIVES may have more serious implications than FALSE POSITIVES in some medical diagnoses

Different Functional Classification Schemes

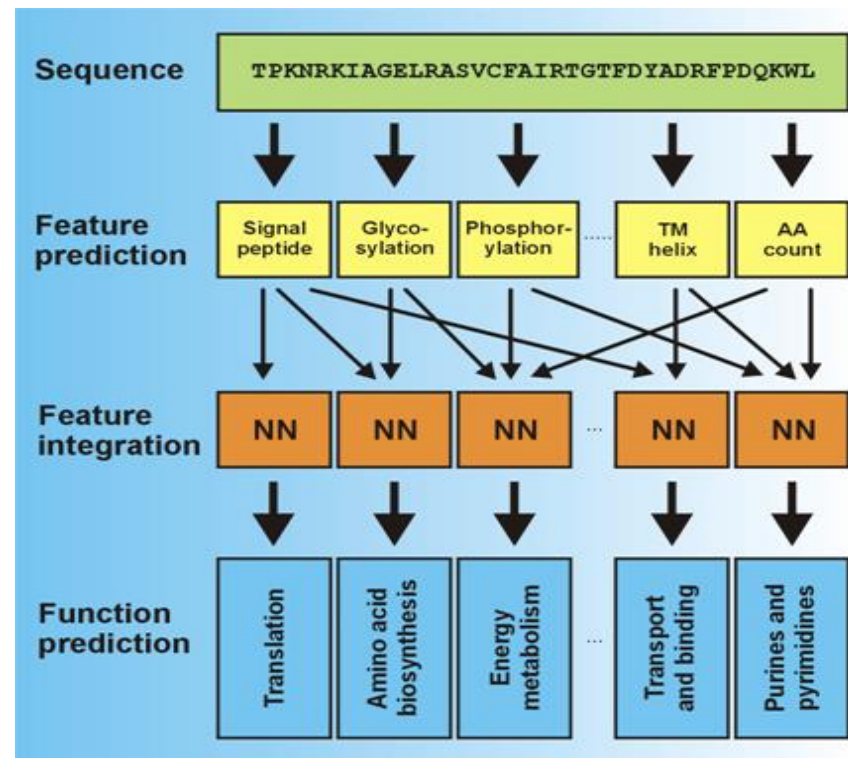
- Functional role (by Monica Riley)
 - Only 14 categories (including “other”) e.g. “transcription”
- Enzyme classification (Enzyme Commission)
 - Hierarchical scheme only applicable to enzymes
 - Some classes have few examples
- Gene Ontology
 - Acyclic graph structure i.e. proteins can belong to multiple classes
 - Some subjectivity in design
 - Some categories have few examples
 - Can select prediction classes by number of examples

- SWISS-PROT can provide a lot of functional information on proteins
- BUT we need to be careful e.g. do we know the *PROVENANCE* of the DATA?
- Extracting the information can require sophisticated parsing e.g. TEXT MINING techniques

```
//
ID 1A1D_PYRAB  STANDARD;   PRT;  330 AA.
AC  Q9V2L2;
DT  30-MAY-2000 (Rel. 39, Created)
DT  30-MAY-2000 (Rel. 39, Last sequence update)
DT  10-OCT-2003 (Rel. 42, Last annotation update)
DE  Putative 1-aminocyclopropane-1-carboxylate deaminase (EC 3.5.99.7)
DE  (ACC deaminase).
GN  PYRAB00630 OR PAB2303.
OS  Pyrococcus abyssi.
OC  Archaea; Euryarchaeota; Thermococci; Thermococcales; Thermococcaceae;
OC  Pyrococcus.
OX  NCBI_TaxID=29292;
RN  [1]
RP  SEQUENCE FROM N.A.
RC  STRAIN=GE5 / Orsay;
RX  MEDLINE=22511545; PubMed=12622808;
RA  Cohen G.N., Barbe V., Flament D., Galperin M., Heilig R., Lecompte O.,
RA  Poch O., Prieur D., Querellou J., Ripp R., Thierry J.-C.,
RA  Van der Oost J., Weissenbach J., Zivanovic Y., Forterre P.;
RT  "An integrated analysis of the genome of the hyperthermophilic
RT  archaeon Pyrococcus abyssi.";
RL  Mol. Microbiol. 47:1495-1512(2003).
CC  -|- CATALYTIC ACTIVITY: 1-aminocyclopropane-1-carboxylate + H(2)O = 2-
CC  oxobutanoate + NH(3).
CC  -|- COFACTOR: Pyridoxal phosphate (By similarity).
CC  -|- SIMILARITY: Belongs to the ACC deaminase/D-cysteine desulphydrase
CC  family.
KW  Hypothetical protein; Hydrolase; Pyridoxal phosphate;
KW  Complete proteome.
FT  BINDING   54   54   Pyridoxal phosphate (By similarity).
SQ  SEQUENCE  330 AA;  35754 MW;  CC8699923C0B11CD CRC64;
```

The concept of ProtFun

- Predict as many biologically relevant features as we can from the sequence
- Train artificial neural networks for each category, also optimizing the feature combinations
- Assign a probability for each category from the NN outputs



Applying neural networks to protein function prediction

- Human protein sequences from SWISS-PROT were assigned to functional classes based on their keywords
- The set of sequences was divided into a test and a training set with no significant sequence similarity between the two sets
- Neural networks were initially trained using individual features and subsequently trained on combinations of the best performing features

Why use derived features?

Why not just use the raw sequence?

MRFLLLAAFVAYAYAKSDEEIRKDALSALDVVPLGSTPEKLENGR
EFYKYFFTNNHQDLRKYFKGAETFTADDIAKSDRFKKLGNQLLLSV
HLAADTYDNEMIFRAFVRDTIDRHVDRGLDPKLWKEFWSIYQKFL
ESKGKTLSADQKAAFDAIGTRFNDEAQKQLAHHGLPHT

ETYTVKLGSDKGLLVFEPAKLTIKPGDTVEFLNNKVPPHNVVFDA
ALNPAKSADLAK

EGREFYKYFFTNNHQDLRKYFKGAGLLVFEPAKLTIKPGDTVEFLNNKVPPHNVVFDAALNPAKSADLAFEPAKLTIKPGDTVEFLNNKVPPHNVVFDAALNPAKSADLAKMRFLLLAAFVAY
AYAKSDEEIRKDALSALDVVPLGSTPEKLENGREFYKYFFTNNHQDLRKYFKGAETFTADDIAKSDRFKKLGNQLLLSVHLAADTYDNEMIFRAFVRDTIDRHVDRGLDPKLWKEFWSIYQKF
LESKKMRFLLAADTYDNEMIFRAFVRDTIDRHVDRGLDPKLWKEFWSIYQKFLESKKMRFLAAFVAYAYAKSDEEIRKDALSALDVVPLGSTPEKLENGREFYKYFFTNNHQDLRKYFKGAGL
LVFEPAKLTIKPGDTPGDTVEFLNNKVPPHNVTYTVKLGSDKGLLVFEPAKLTIKPGDTVEFLNNKVPPHNVVFDAALNPAKSADLAFEPAKLTIKPGDTVEFLNNKVPPHNVVFDA
ALNPAKSADLAKMRFLPHNVVFDAALNPAKSADLAKMRFLLLAAFVAYAYAKSHNVVFDAALNPAKSADLAFEPAKLTIKPGDTVEFLNNKVPPHNVVFDAALNPAKSADLAKMRFLLLAA
FVAYAYAKSDEEIRKDALSALDVVPLGSTPEKLENGREFYKYFFTNNHQDLRKYFKGAETFTADDIAKSDRFKKLGNQLLLSVHLGSTPEKLENGREFYKYFFTNNHQDLRKYFKGAGLLVFEP
AKLTIKPGDTVEFLNNKVPPVAYAYAKSDEEIRKDALSALDVVPLGSTPEKLENVFDAALNPAKSADLAKMRFLLLAAFVAYAYAKSDEEIRKLAAFVAYAYAKSDEEIRKDALSALDVVPL
GSTPEKLENGREFYKYFFTNNHQDLRKYFKGAETFTADDIAKSDRFKKLGNQLLLSVHLAADTYDNEMIFRAFVRDTIDRHVDRGLDPKLWKEFWSIYQKFLESKKMRFLLLAAFVAYAYAKS
DEEIRKDALSALDVVPLGSTPEKLENGREETFTADDIAKSDRFKKLGNQLLLSVHLAADTYDNEMIFRAFVRDTIDRHVDRGLDPKLWKEFWSIYQKFLESKKMRFLLDALSALDVVPLLAA
FVAYAYAKSDEEIRKDALSALDVHNVVFDAALNPAKSADLAFEPAKLTIKPGDTVEFLNVPLGSTPEKLENVFLNNKVPPHNVVFDAALNPAKSADLAFEPAKLTIKPGDTVEFLNNKVPL
LLAAFVAYAYAKSDEEIRKDALSALDVVPLGSTPEKLENVFLNNKVPPHNVVFDAALNPAKSADLAFEPAKLTIKPGDTVEFLNNKVPPHNVVFDAALNPAKSADLAKMRFLLLAAFVAYA
YAKSDEEIRKDALSALDVVPLGSTPEKLENGREFYKYFFTNNHQDLRKYFKGAETFTADDIAKSDRFKKLGNQLLLSVHLAADTYDNEMIFRAFVRDTIDRHVDRGLDPKLWKEFWSIYQKFL
ESKGKTLSADQKAAFDAIGTRFNDEAQKQLAHHGLPHT

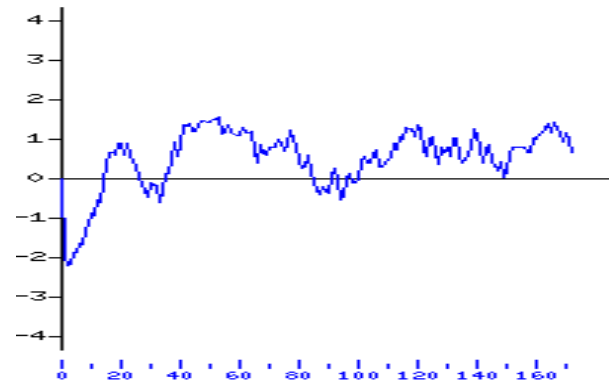
Example of a Feature

Hydrophobicity Scales $H(x)$

Kyte-Doolittle Hopp-Woods

Alanine	1.8	-0.5
Arginine	-4.5	3.0
Asparagine	-3.5	0.2
Aspartic acid	-3.5	3.0
Cysteine	2.5	-1.0
Glutamine	-3.5	0.2
Glutamic acid	-3.5	3.0
Glycine	-0.4	0.0
Histidine	-3.2	-0.5
Isoleucine	4.5	-1.8
Leucine	3.8	-1.8
Lysine	-3.9	3.0
Methionine	1.9	-1.3
Phenylalanine	2.8	-2.5
Proline	-1.6	0.0
Serine	-0.8	0.3
Threonine	-0.7	-0.4
Tryptophan	-0.9	-3.4
Tyrosine	-1.3	-2.3
Valine	4.2	-1.5

MRFLLLAAFVAYAYAKSDEEIRKDALSDVVP LGSTPEKLENGREFYKY
 FFTNHQDLRK YFKGAETFTADDIAKSDRFKKLGNQLLLSVHLAADTYDNE
 MIFRAFVRDTIDRHVDRGLDPKLWKEFWSIYQKFLESKGKTL SADQKA AF
 DAIGTRFNDEAQKQLAHHGLPHT



$$\frac{1}{n} \sum_{i=1}^n H(x_i)$$

What sequence features can we look at?

There are an infinite variety of features that we can come up with!

Examples of ProtFun features:

- Global feature e.g.
 - Average hydrophobicity
 - Number of negatively charged residues
 - Number of positively charged residues
 - Subcellular location
- Site features e.g.
 - Glycosylation sites
 - Signal peptides
- Structural features e.g.
 - Secondary structure
 - Transmembrane helices

```
MRFLLLAAFVAYAYAKSDEE  
IRKDALSALDVVPLGSTPEK  
LENGREFYKYFFTNHQDLRK  
YFKGAETFTADDI AKSDRFK  
KLGNQLLLSVHLAADTYDNE  
MIFRAFVRDTIDRHVDRGLD  
PKLWKEFWSIYQKFLESKGK  
TLSADQKAAFDAIGTRFNDE  
AQKQLAHHGLPHT
```

Feature Encoding

How should each feature be encoded?

- We want a fixed width training vector
 - May need to using pad shorter vectors
- Numeric ranges of feature values may differ greatly
 - May need normalisation or scaling function
- Some features may be absent or invalid for some samples
 - May need an input which represents “not known”
- Feature encoding may be critical to success
 - Need to benchmark alternative encoding schemes

	EVENTS	NEWS	RESEARCH GROUPS	CBS PREDICTION SERVERS	CBS DATA SETS
	STAFF	CONTACT	ABOUT CBS	INTERNAL	CBS BIOINFORMATICS TOOLS

[CBS](#) >> [CBS Prediction Servers](#) >> ProtFun

ProtFun 2.2 Server

Prediction of cellular role, enzyme class and Gene Ontology category

The ProtFun 2.2 server produces *ab initio* predictions of protein function from sequence. The method queries a large number of other feature prediction servers to predict the cellular role, enzyme class (if any), and selected Gene Ontology categories of the submitted sequence.

It is possible to inspect the individual feature predictions used and integrated by ProtFun.

View the [version history](#) of this server.

[Instructions](#)

[Output format](#)

SUBMISSION

Paste a single sequence or several sequences in [FASTA](#) format into the field below:

Submit a file in [FASTA](#) format directly from your local disk:

No file chosen

Show the individual features (*PTM*:s, sorting signals etc.) in the output ☐

Show the prediction process as it progresses ☐

MRFLLLAAFVAYAYAKSDEEIRKDALSALDVVPLGSTPEKLE
NGREFYKYFFTNNHQDLRKYFKGAETFTADDIAKSDRFKLG
QLLSVHLAADTYDNEMIFRAFVRDTIDRHVDRGLDPKLWK
EFWSIYQKFLESKGKTL SADQKA AFDAIGTRFNDEA QKQLA
HHGLPHT

Feature	Output summary	Full output
SignalP 3.0	Most likely cleavage site between pos. 15 and 16: AYA-KS	details
ProP 1.0	No polypeptide cleavage sites predicted	details
TargetP 1.01	Secretory signal peptide predicted	details
NetPhos 2.0	8 putative phosphorylation sites at positions 17 27 76 37 96 110 155 97	details
NetOGlyc 3.1	1 putative O-glycosylated site at position 173	details
NetNGlyc 1.0	No N-glycosylated sites predicted	details
TMHMM 2.0	No TM helices predicted	details

ProtFun 2.2 predictions for TestSeq

>TestSeq

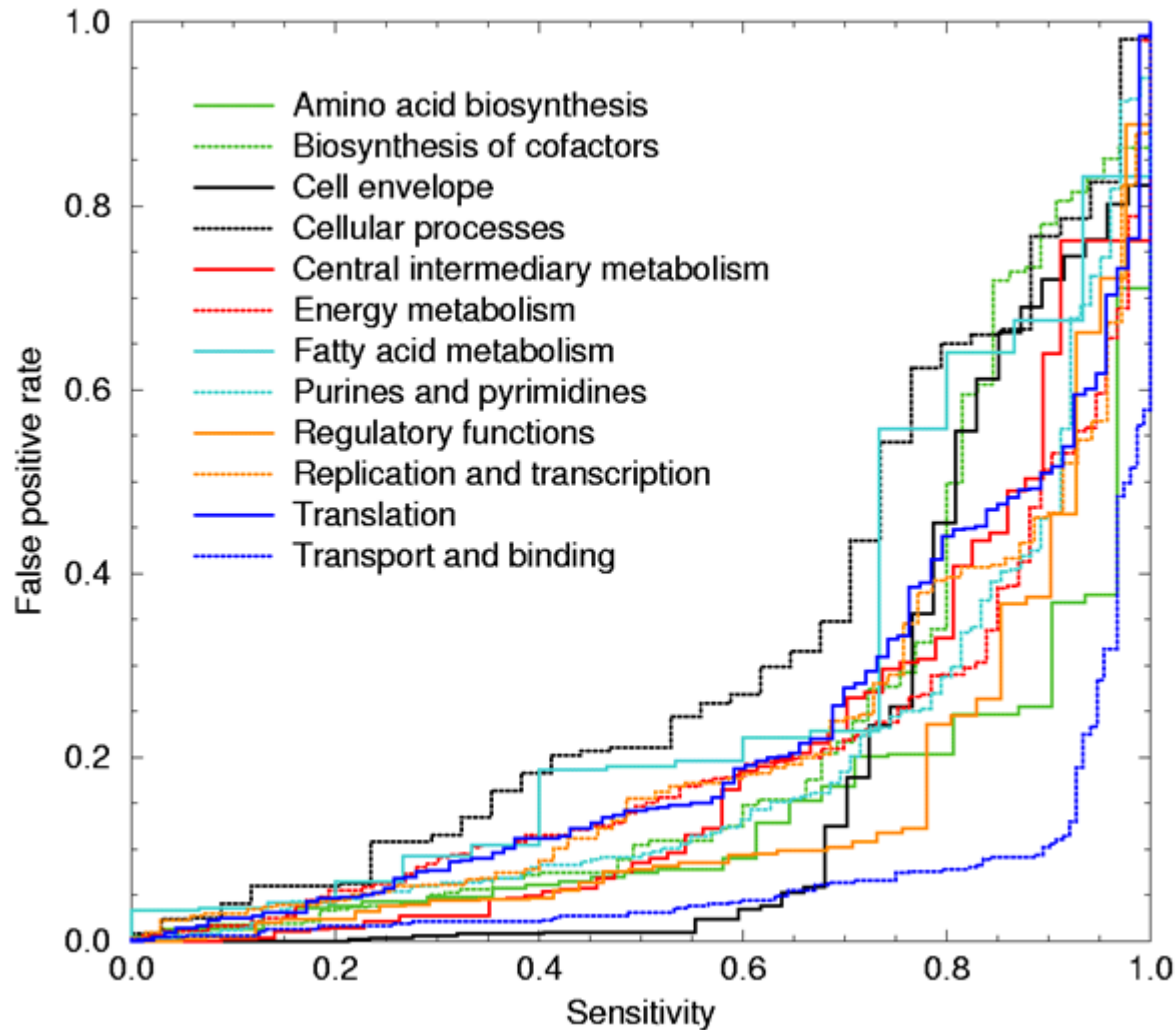
# Functional category	Prob	Odds
Amino_acid_biosynthesis	0.043	1.943
Biosynthesis_of_cofactors	0.079	1.102
Cell_envelope	=> 0.701	11.488
Cellular_processes	0.070	0.957
Central_intermediary_metabolism	0.042	0.660
Energy_metabolism	0.031	0.342
Fatty_acid_metabolism	0.017	1.328
Purines_and_pyrimidines	0.265	1.090
Regulatory_functions	0.013	0.080
Replication_and_transcription	0.020	0.073
Translation	0.035	0.794
Transport_and_binding	0.827	2.018
# Enzyme/nonenzyme	Prob	Odds
Enzyme	=> 0.337	1.176
Nonenzyme	0.663	0.929
# Enzyme class	Prob	Odds
Oxidoreductase (EC 1.-.-.-)	0.054	0.258
Transferase (EC 2.-.-.-)	0.039	0.113
Hydrolase (EC 3.-.-.-)	0.079	0.250
Lyase (EC 4.-.-.-)	0.043	0.919
Isomerase (EC 5.-.-.-)	0.010	0.321
Ligase (EC 6.-.-.-)	0.017	0.326
# Gene Ontology category	Prob	Odds
Signal_transducer	0.195	0.912
Receptor	0.006	0.035
Hormone	0.004	0.542
Structural_protein	0.001	0.046
Transporter	0.025	0.232
Ion_channel	0.013	0.220
Voltage-gated_ion_channel	0.004	0.190
Cation_channel	0.010	0.215
Transcription	0.029	0.229
Transcription_regulation	0.021	0.171
Stress_response	=> 0.211	2.402
Immune_response	0.018	0.217
Growth_factor	0.012	0.829
Metal_ion_transport	0.009	0.020

IS THIS CORRECT?

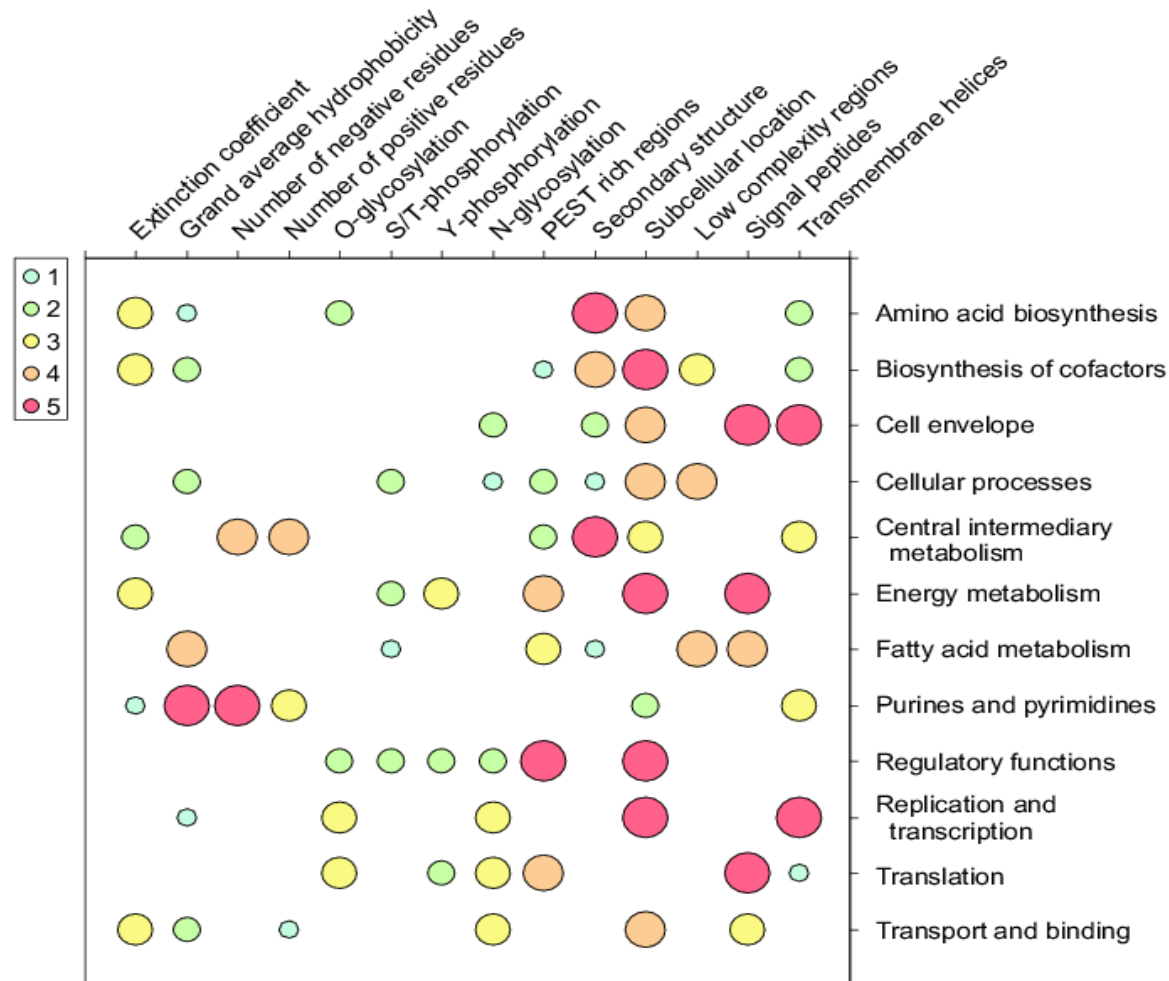
Try checking this result yourself. The target sequence was that of human myoglobin. You can easily look it up in Uniprot (<http://www.uniprot.org>) and check the given keywords or Gene Ontology classification.

In this case, would it have been better to just look at the raw probabilities and ignore the priors?

Prediction performance on cellular role categories

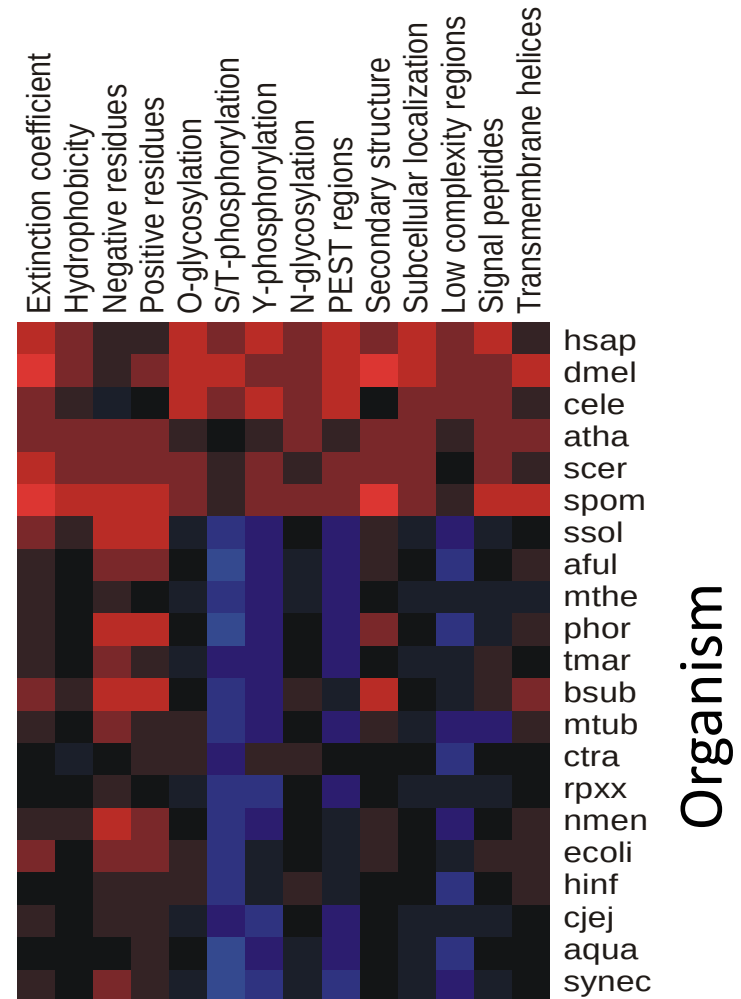


Which features are most useful?



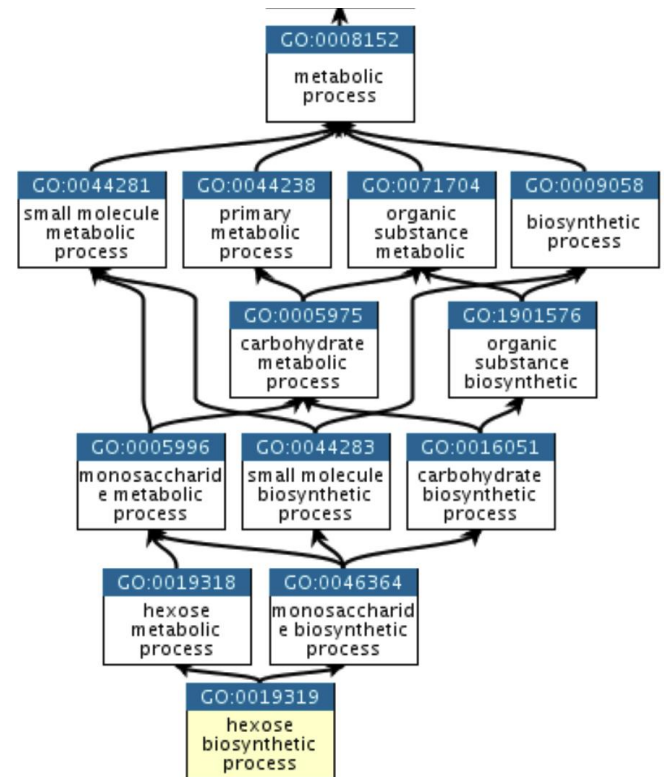
Performance contribution of sequence derived features

- Importance of sequence features can depend on the source of the protein i.e. which organism it derives from
- No variations are observed across different eukaryotes
- However some features are most useful for archaea and bacteria
 - Physical/chemical properties
 - Secondary structure and transmembrane helices
- Some features only apply to eukaryotes
 - E.g. Subcellular localization features



A better classification system: the Gene Ontology

- The Gene Ontology (see earlier) provides a much better set of classifications
- Acyclic graph structure i.e. proteins can belong to multiple classes
- Different aspects of function are represented in GO:
 - Molecular function
 - Biological process
 - Cellular component
- We can choose an arbitrary level of detail i.e. decide how far down the graph we will descend
- Generally we limit ourselves to training only on *experimentally validated* GO terms
- In Swiss-Prot (Jun 2024) there are 28,447 different GO term labels, however these occur with very different frequencies

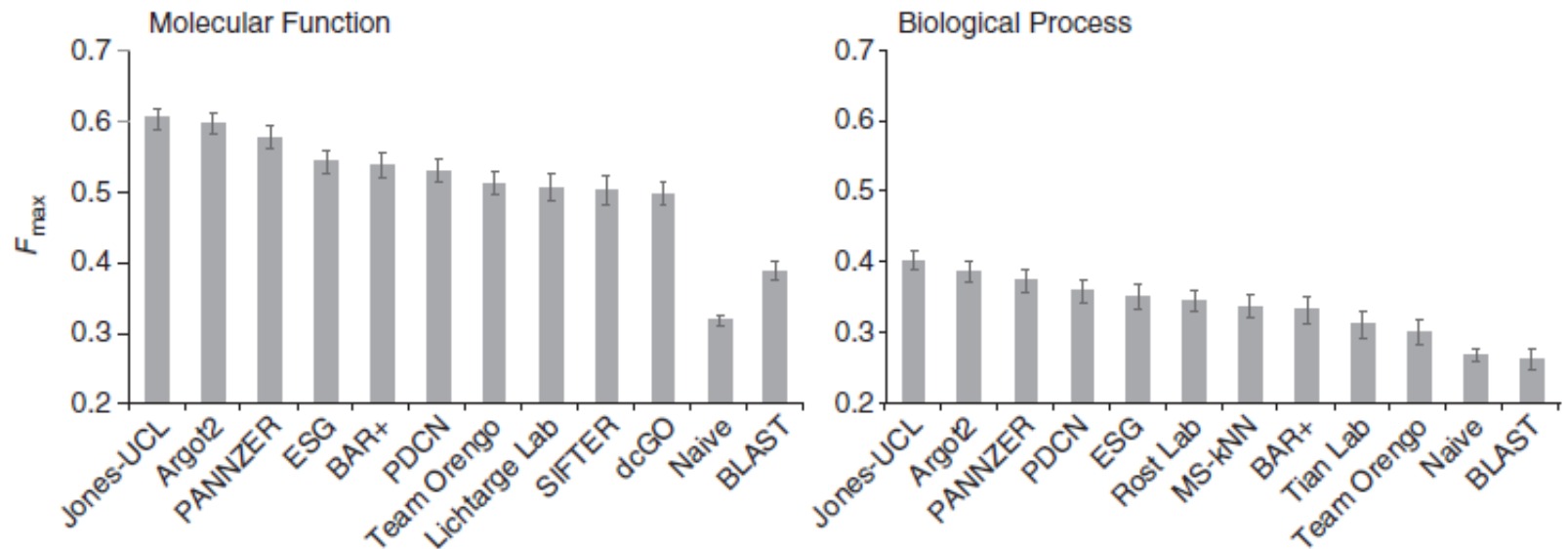


Swiss-Prot GO Term Data

(Terms ranked by frequency)

```
1 GO:0005634 20312 cellular_component: nucleus
2 GO:0005737 20071 cellular_component: cytoplasm
3 GO:0005829 15892 cellular_component: cytosol
4 GO:0005886 15563 cellular_component: plasma membrane
5 GO:0016020 8810 cellular_component: membrane
6 GO:0046872 8601 molecular_function: metal ion binding
7 GO:0005524 8074 molecular_function: ATP binding
8 GO:0005654 7116 cellular_component: nucleoplasm
9 GO:0042802 5814 molecular_function: identical protein binding
10 GO:0005739 5676 cellular_component: mitochondrion
11 GO:0005576 5604 cellular_component: extracellular region
12 GO:0003677 4543 molecular_function: DNA binding
13 GO:0005615 4282 cellular_component: extracellular space
14 GO:0005783 3980 cellular_component: endoplasmic reticulum
15 GO:0006357 3950 biological_process: regulation of transcription by RNA polymerase II
...
1534 GO:0150104 100 biological_process: transport across blood-brain barrier
1535 GO:0061512 100 biological_process: protein localization to cilium
1536 GO:0051233 100 cellular_component: spindle midzone
1537 GO:0050775 100 biological_process: positive regulation of dendrite morphogenesis
...
28447 GO:0000024 1 biological_process: maltose biosynthetic process
```

Function Prediction from Sequence Remains a Hard Problem



From: A large-scale evaluation of computational protein function prediction. Radivojac et al. (2013) Nature Methods. 10, 221–227.
doi:10.1038/nmeth.2340

Some “winning” Function Prediction Strategies

- Learning-to-rank methods work well
- Combining text-mining, distant homology and protein Language Model (pLM) embeddings
- Neural network architectures modelled on the GO acyclic graph structure
- Other data sources such as transcriptomics or proteomics data can be useful, but are species-specific

Optional further reading...

Predicting human protein function with multi-task deep neural networks (2018)

Rui Fa, Domenico Cozzetto, Cen Wan, David T. Jones

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5995439>

DeepGOZero: improving protein function prediction from sequence and zero-shot learning based on ontology axioms (2022)

Maxat Kulmanov, Robert Hoehndorf

https://academic.oup.com/bioinformatics/article/38/Supplement_1/i238/6617515

CAFA 5 Protein Function Prediction Kaggle:

<https://www.kaggle.com/competitions/cafa-5-protein-function-prediction>